Assessment 1 — Solutions

Every question reproduced exactly, followed by a step-by-step answer, circuit, working table and waveforms.

10 questions · worked solutions · compiled 21 October 2026

Symbols	V_m peak (secondary) voltage · I_m peak load current · R_L load resistance · r_f diode forward resistance
Silicon diode	Cut-in / knee voltage $V_\gamma = V_T = 0.7$ V (ideal diode: 0 V)
Handy numbers	$1/\pi = 0.318$ · $2/\pi = 0.636$ · $\sqrt{2} = 1.414$ · $1/\sqrt{2} = 0.707$
Colours	blue = input / main waveform · vermilion = rectified output or current path · grey dashed = reference / non-conducting part

Exam recap

Always (1) convert rms → peak with √2, (2) apply the turns ratio, (3) subtract 0.7 V per conducting Si diode (one for HWR/CT, two for bridge), (4) use 0.318 or 0.636, (5) PIV: V_m (HWR, bridge) or $2V_m$ (CT).

Contents

1. Block diagram of a regulated power supply
2. Half-wave rectifier: circuit and working table
3. Negative half-wave rectifier, ideal vs silicon, 20 V vs 200 V
4. Full-wave centre-tapped rectifier: circuit and working table
5. Full-wave bridge rectifier: circuit and working table
6. Formula comparison table: HWR, FWR-CT, Bridge
7. Bridge rectifier, 120 V rms, 1 kΩ, silicon diodes
8. Two-diode FWR, 100 V per half, 975 Ω
9. Bridge rectifier, 220 V, 4 : 1 transformer, 200 Ω
10. Bridge rectifier, 230 sin 314t, 10 : 1, 500 Ω

Q1 Regulated power supply — block diagram

> Question 1
>
> Write block diagram of Regulated power supply with each block showing simple input and output graph.

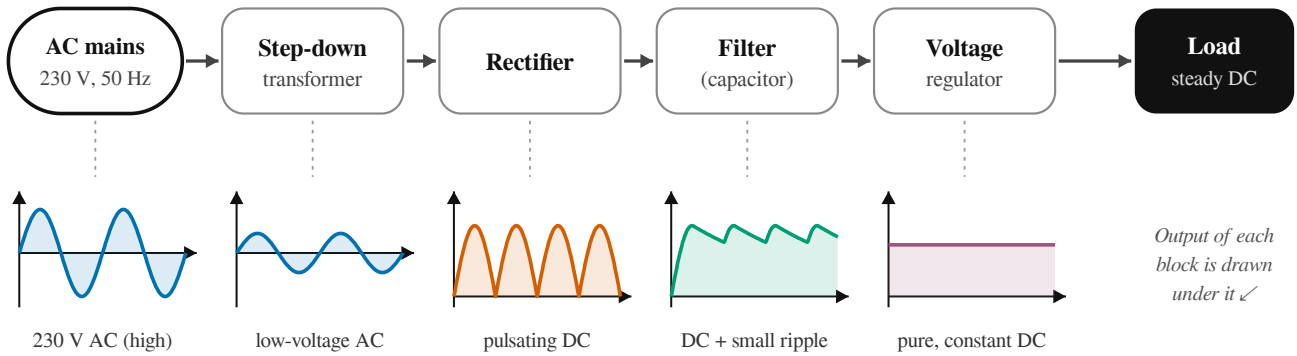


Figure 1 · Block diagram of a regulated DC power supply. The small graph under each block is that block's **output**, which is also the **input** of the next block.

Read left to right: the wave starts as high-voltage AC and ends as a steady DC level.

Block	Input	Output	Function
Step-down transformer	230 V, 50 Hz AC (large sine)	Low-voltage AC, same 50 Hz (small sine)	Reduces the mains voltage to the level needed (e.g. 12 V); also isolates the circuit from the mains.
Rectifier	Low-voltage AC	Pulsating DC (unidirectional humps)	Diodes conduct in one direction only, so the negative halves are removed (HWR) or flipped (FWR).
Filter	Pulsating DC	DC with small ripple	A capacitor charges to the peak and supplies the load between peaks, smoothing the humps.
Voltage regulator	DC with ripple (varies with mains and load)	Pure, constant DC	Zener diode or IC (78xx) holds the output fixed in spite of input or load changes.
Load	Constant DC	—	The circuit being powered (e.g. an amplifier, a phone charger).

Remember the order: *T*ransformer → *R*ectifier → *F*ilter → *R*egulator → Load ("TRFR"). An *unregulated* supply is the same chain without the regulator block.

Q2 Half-wave rectifier — circuit and working

Question 2

Write a circuit diagram of Half Wave Rectifier and write it's working in the table format. For ex.

Cycle | Diode(on/off) | Current | Vout graph(for one full cycle)

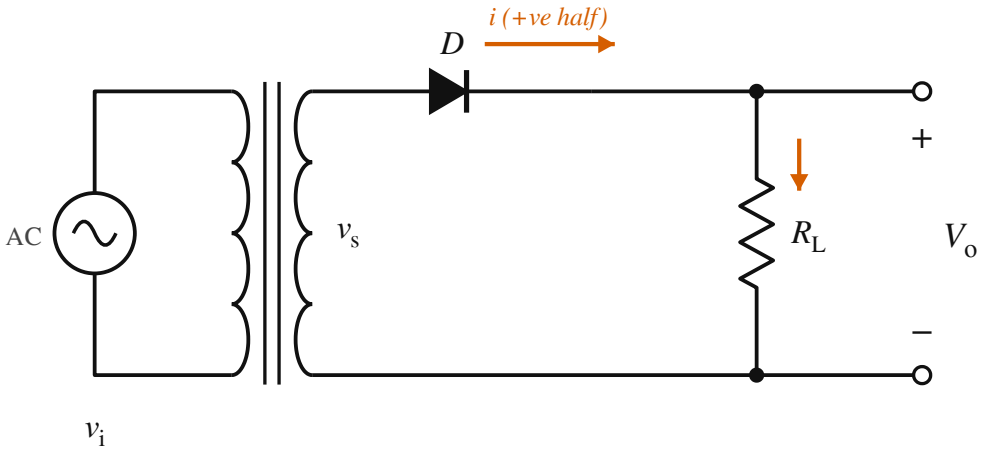
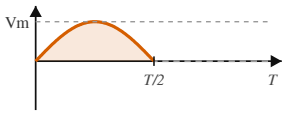
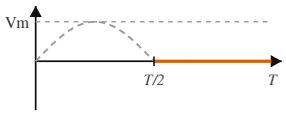


Figure 2.1 · Half-wave rectifier: transformer secondary, one diode D in series with the load R_L.

Vermilion arrows: direction of current during the positive half-cycle.

Cycle	Diode(s) ON / OFF	Current path	V_{out} graph (one full cycle)
Positive half $0 \rightarrow T/2$ top of secondary +	ON forward biased (anode +)	Flows: secondary (+) → D → R_L (top to bottom) → back to secondary. $i = v_s/R_L$	 $V_{out} = v_s = V_m \sin \omega t$
Negative half $T/2 \rightarrow T$ top of secondary –	OFF reverse biased (anode –)	No current: diode acts as an open switch. $i = 0$	 $V_{out} = 0$; diode blocks PIV = V_m

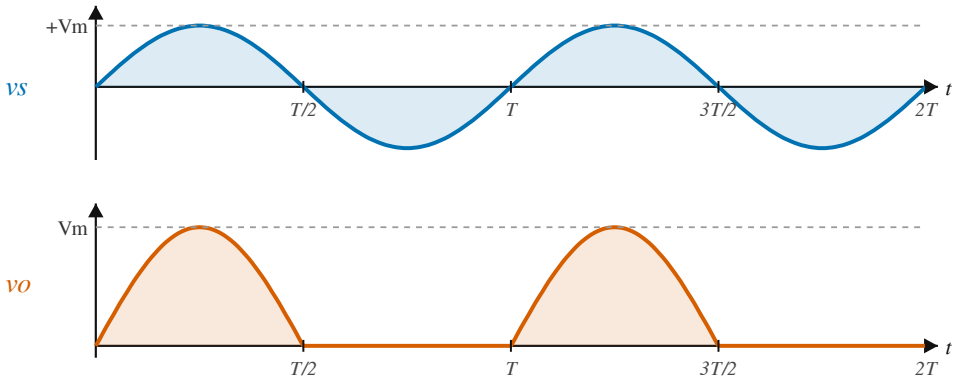


Figure 2.2 · Input and output over two cycles: only the positive halves reach the load.

Average output $V_{dc} = V_m/\pi = 0.318\ V_m$; output frequency = input frequency (50 Hz).

Q3 Sketch v_o and find the dc level

> Question 3
>
> a. Sketch the output v_o and determine the dc level of the output for the network given.
>
> b. Repeat part (a) if the ideal diode is replaced by a silicon diode.
>
> c. Repeat parts (a) and (b) if V_m is increased to 200 V, and compare solutions.
>
> Network: $v_i = 20$ V peak sinusoid (period T), diode in series, $R = 2$ kΩ to ground, output across R.

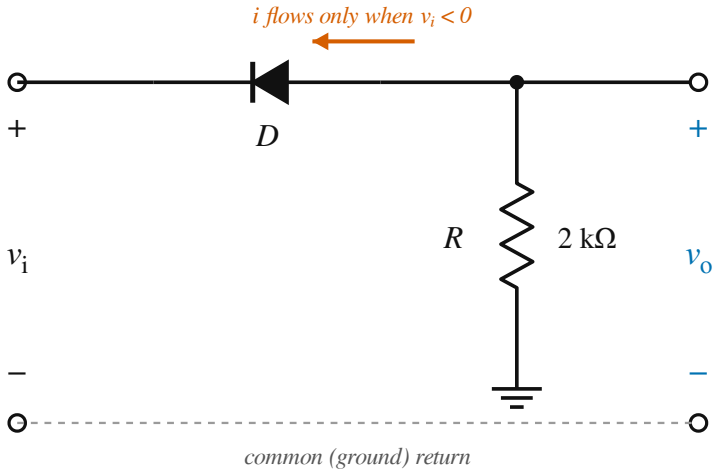


Figure 3.1 · The given network. The diode's **anode faces the output** (bar towards v_i).

So the diode is forward biased only when v_i is negative: this is a **negative** half-wave rectifier.

1 **Decide when the diode conducts**

Anode is at the resistor side (≈ 0 V through R to ground), cathode at v_i. Diode is ON when cathode is below anode, i.e. when $v_i < 0$ (second half, $T/2 \to T$). For $v_i > 0$ it is OFF and $v_o = 0$.

2 **(a) Ideal diode: output and dc level**

During the negative half the ideal diode is a short, so $v_o = v_i$: a negative pulse of peak −20 V.

$V_{dc} = -0.318\ V_m = -0.318 \times 20 = \mathbf{-6.36\ V}$

3 **(b) Silicon diode: subtract 0.7 V from the peak**

The diode needs 0.7 V before it conducts, so the output peak is 0.7 V smaller in magnitude.

$V_m' = -(20 - 0.7) = -19.3\ \text{V} \Rightarrow V_{dc} = -0.318 \times 19.3 = \mathbf{-6.14\ V}$

4 **(c) Repeat with V_m = 200 V**

Ideal: $V_{dc} = -0.318 \times 200 = \mathbf{-63.6\ V}$

Silicon: $V_{dc} = -0.318 \times (200 - 0.7) = -0.318 \times 199.3 = \mathbf{-63.38\ V}$

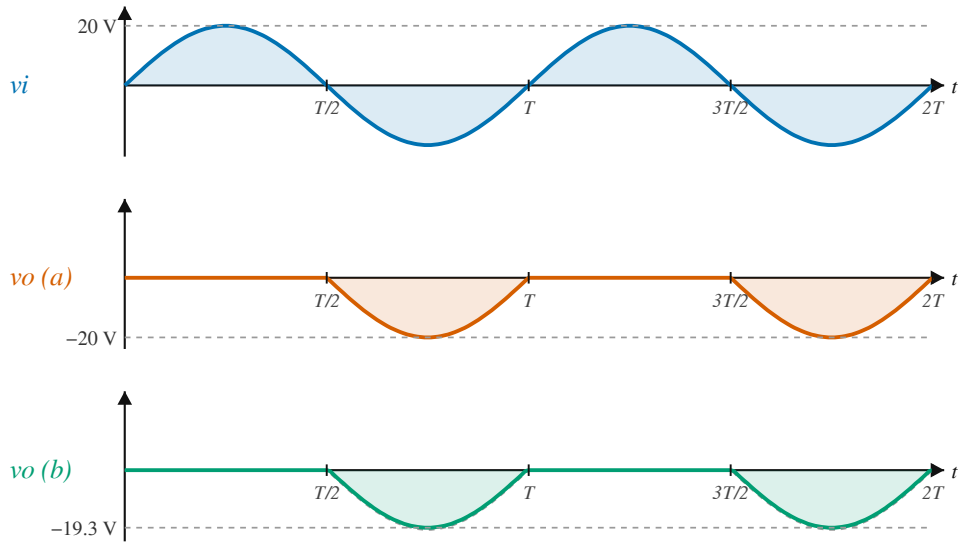


Figure 3.2 · Sketch of v_o: (a) ideal diode, negative pulses of −20 V; (b) silicon diode, pulses of −19.3 V that start slightly late and end slightly early (grey dashed = ideal).

In (b) the diode stays OFF while $|v_i| < 0.7$ V, so each pulse is a little narrower.

Case	V_m = 20 V	V_m = 200 V
Ideal diode	−6.36 V	−63.6 V
Silicon diode	−6.14 V	−63.38 V
Difference	0.22 V (≈ 3.5 %)	0.22 V (≈ 0.35 %)

Answer

Comparison

The 0.7 V drop always costs the same ≈ 0.22 V of dc level, but at 20 V it is a noticeable 3.5 % error while at 200 V it is only 0.35 %. **For large peak voltages ($V_m \gg V_T$) the ideal-diode model is accurate enough;** for small voltages the 0.7 V must be included.

Q4 Full-wave centre-tap rectifier — circuit and working

Question 4

Write a circuit diagram of FWR-CT and it's working in table format.

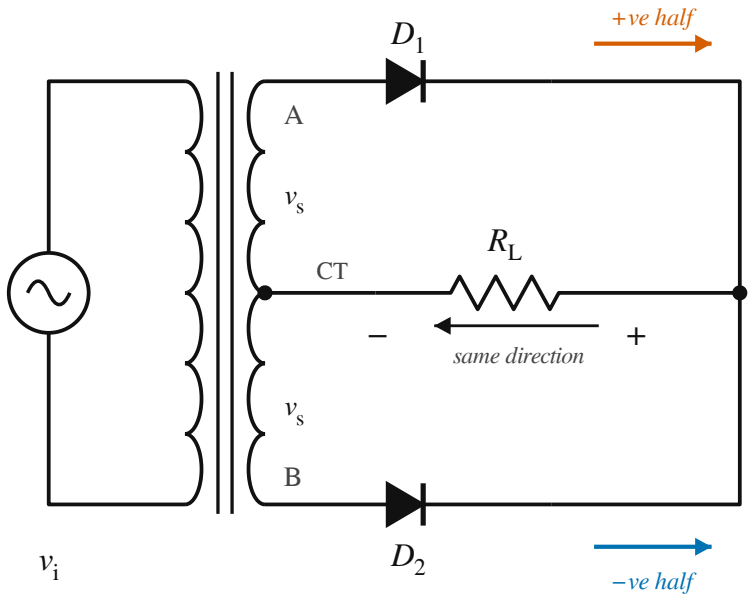
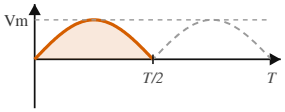
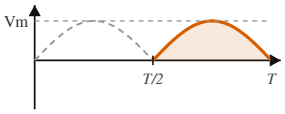


Figure 4.1 · Full-wave centre-tapped rectifier: two diodes, each fed by one half of a centre-tapped secondary.

Each half of the secondary gives v_s (peak V_m). Current through R_L is in the same direction in both halves.

Cycle	Diode(s) ON / OFF	Current path	V_{out} graph (one full cycle)
Positive half $0 \to T/2$ A +, B – w.r.t. CT	D1 ON D2 OFF	A → D1 → R_L (+ to –) → centre tap. $i = v_s/R_L$	 $V_{out} = V_m \sin \omega t$
Negative half $T/2 \to T$ B +, A – w.r.t. CT	D1 OFF D2 ON	B → D2 → R_L (+ to –, **same direction**) → centre tap. $i = \lvert v_s \rvert / R_L$	 $V_{out} = \lvert V_m \sin \omega t \rvert$; OFF diode sees PIV = $2V_m$

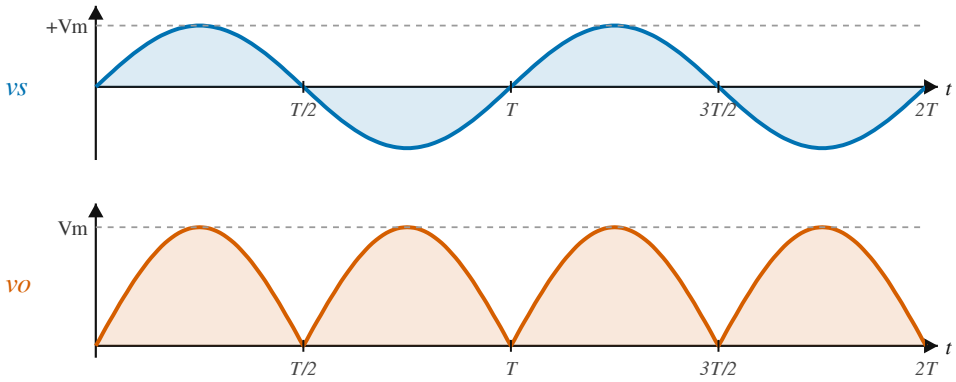


Figure 4.2 · Both halves of the input appear across the load with the same polarity.

$V_{dc} = 2V_m/\pi = 0.636\ V_m$; output ripple frequency = 2 × 50 = 100 Hz.

Q5 Full-wave bridge rectifier — circuit and working

> Question 5
>
> Write a circuit diagram of FWR-Bridge and it's working in the table format.

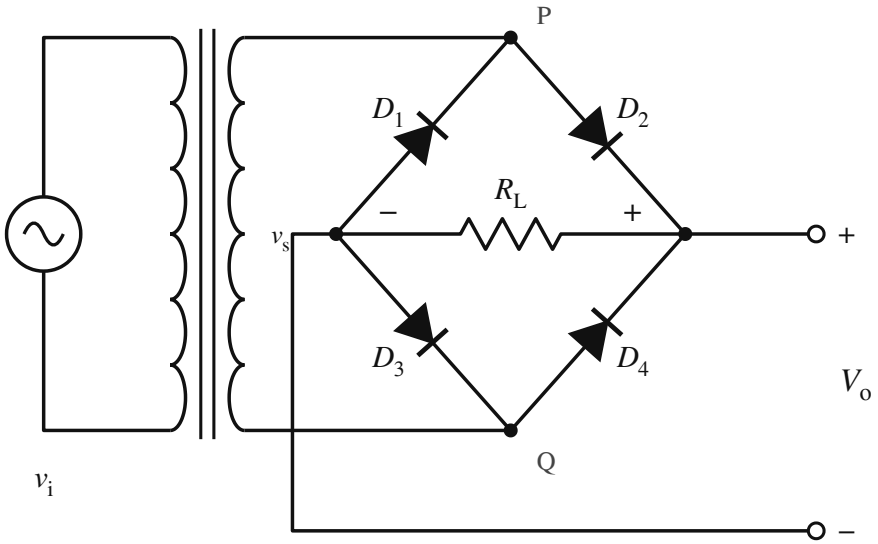
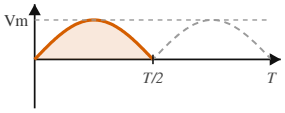
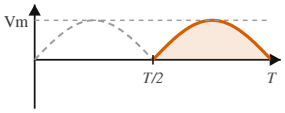


Figure 5.1 · Bridge rectifier: four diodes D1–D4 in a bridge; AC applied across P–Q, load across the other two corners.

The cathodes of D2 and D4 meet at the + output; the anodes of D1 and D3 meet at the − output.

Cycle	Diode(s) ON / OFF	Current path	V_{out} graph (one full cycle)
Positive half 0 → *T*/2 P +, Q −	D2, D3 ON D1, D4 OFF	P → D2 → R_L (+ to −) → D3 → Q. Two diodes in series with the load.	 $V_{out} = V_m \sin \omega t$

Cycle	Diode(s) ON / OFF	Current path	V_{out} graph (one full cycle)
Negative half $T/2 \rightarrow T$ Q +, P –	D4, D1 ON D2, D3 OFF	Q → D4 → R_L (+ to –, **same direction**) → D1 → P.	 $V_{out} = \lvert V_m \sin \omega t \rvert$; each OFF diode sees PIV = V_m

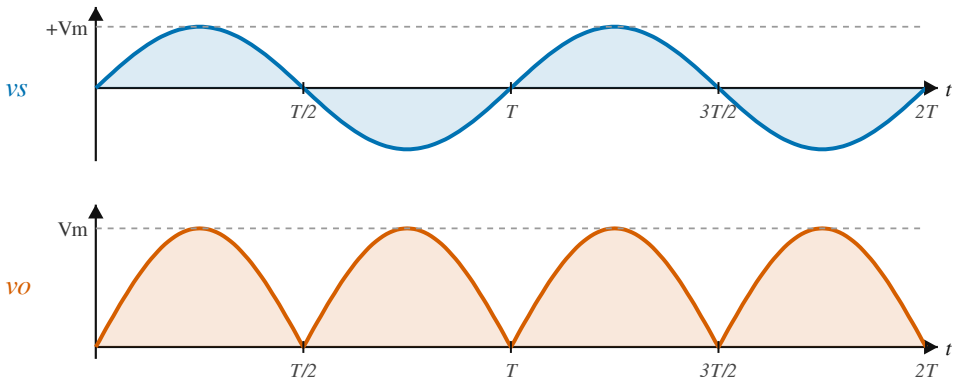


Figure 5.2 · Output identical to the centre-tap circuit, but no centre-tapped transformer is needed.

$V_{dc} = 0.636\ V_m$ (ideal) or $0.636(V_m - 1.4\ V)$ for silicon, because two diodes conduct at a time.

Q6 Formula comparison table

> Question 6
>
> Write formula comparison table for HWR and FWR(CT and Bridge) for both ideal and non-ideal case. (Parameters: Vdc, Idc, Vrms, Irms, PIV)

(a) Ideal diode ($V_\gamma = 0$, $r_f = 0$)

Parameter	HWR	FWR – centre tap	FWR – bridge
Peak current I_m	$\frac{V_m}{R_L}$	$\frac{V_m}{R_L}$	$\frac{V_m}{R_L}$
V_{dc}	$\frac{V_m}{\pi} = 0.318\ V_m$	$\frac{2V_m}{\pi} = 0.636\ V_m$	$\frac{2V_m}{\pi} = 0.636\ V_m$
I_{dc}	$\frac{I_m}{\pi}$	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
V_{rms}	$\frac{V_m}{2}$	$\frac{V_m}{\sqrt{2}}$	$\frac{V_m}{\sqrt{2}}$
I_{rms}	$\frac{I_m}{2}$	$\frac{I_m}{\sqrt{2}}$	$\frac{I_m}{\sqrt{2}}$
PIV	V_m	$2V_m$	V_m

(b) Practical / non-ideal diode (cut-in $V_\gamma = 0.7$ V for Si, forward resistance r_f)

Parameter	HWR	FWR – centre tap	FWR – bridge
Peak current I_m	$\frac{V_m - V_\gamma}{R_L + r_f}$	$\frac{V_m - V_\gamma}{R_L + r_f}$	$\frac{V_m - 2V_\gamma}{R_L + 2r_f}$
V_{dc}	$I_{dc}R_L = \frac{I_m R_L}{\pi}$ (r_f=0: $0.318(V_m - V_\gamma)$)	$I_{dc}R_L = \frac{2I_m R_L}{\pi}$ (r_f=0: $0.636(V_m - V_\gamma)$)	$I_{dc}R_L = \frac{2I_m R_L}{\pi}$ (r_f=0: $0.636(V_m - 2V_\gamma)$)
I_{dc}	$\frac{I_m}{\pi}$	$\frac{2I_m}{\pi}$	$\frac{2I_m}{\pi}$
V_{rms}	$I_{rms}R_L = \frac{I_m R_L}{2}$	$I_{rms}R_L = \frac{I_m R_L}{\sqrt{2}}$	$I_{rms}R_L = \frac{I_m R_L}{\sqrt{2}}$
I_{rms}	$\frac{I_m}{2}$	$\frac{I_m}{\sqrt{2}}$	$\frac{I_m}{\sqrt{2}}$
PIV	V_m	$2V_m - V_\gamma$	$V_m - V_\gamma$

(c) Extra comparison points (useful for 2-mark questions)

Parameter	HWR	FWR – centre tap	FWR – bridge
No. of diodes	1	2	4
Diodes conducting per half	1 (one half only)	1	2 in series
Ripple factor γ	1.21	0.482	0.482
Max. efficiency η	40.6 %	81.2 %	81.2 %
Output (ripple) frequency	f (50 Hz)	$2f$ (100 Hz)	$2f$ (100 Hz)
Transformer	ordinary	centre-tapped	ordinary (or none)

Note on V_m: for the centre-tap circuit V_m is the peak of *one half* of the secondary, so the whole secondary has peak $2V_m$ — that is why its PIV is $2V_m$. In the bridge, V_m is the peak of the whole secondary.

Q7 Bridge rectifier, 120 V rms, 1 kΩ, silicon

Question 7

A full-wave bridge rectifier with a 120-V rms sinusoidal input has a load resistor of 1 kΩ.
(a) If silicon diodes are employed, what is the dc voltage available at the load?
(b) Determine the required PIV rating of each diode.
(c) Find the maximum current through each diode during conduction.
(d) What is the required power rating of each diode?

Given

Bridge rectifier, silicon diodes

$V_{rms} = 120$ V · $R_L = 1$ kΩ · $V_T = 0.7$ V per diode, two diodes conduct at a time

1 **Peak input voltage**

$V_m = \sqrt{2} \times V_{rms} = 1.414 \times 120 =$ **169.7 V**

2 **(a) DC load voltage**

Two silicon diodes are in the conduction path, so the load peak is reduced by 2 × 0.7 V.

$V_{Lm} = 169.7 - 2(0.7) = 168.3$ V

$V_{dc} = 0.636 \times 168.3 =$ **107.0 V**

3 **(b) PIV of each diode**

The OFF diode is across the secondary less the drop of the one ON diode beside it; rate it at least V_m.

PIV = V_m − 0.7 = 169.0 V ⇒ choose PIV rating ≥ **169.7 V** (≈ 170 V)

↓

4 **(c) Maximum diode current**

Each conducting diode carries the full load current; its peak is

$$I_{D(max)} = \frac{168.3\ \text{V}}{1\ \text{k}\Omega} = \mathbf{168.3\ mA}$$

↓

5 **(d) Power rating of each diode**

Diode drop × maximum diode current:

$P_{max} = V_T \cdot I_{D(max)} = 0.7 \times 168.3\ \text{mA} = \mathbf{117.8\ mW}$

↓

ANSWER

Results

(a) V_{dc} ≈ 107 V · (b) PIV ≥ 169.7 V · (c) $I_{D(max)}$ = 168.3 mA · (d) P ≈ 117.8 mW

Average power check (d): each diode conducts only every other half-cycle, so its average current is $I_{dc}/2$ = 107.0 mA/2 = 53.5 mA and its average power ≈ 0.7 × 53.5 mA = 37.5 mW. The 117.8 mW figure is the safe (peak) rating normally asked for.

Q8 Two-diode FWR, 100 V per half, 975 Ω

QUESTION 8

In a two diode FWR using Si diodes, the RMS voltage across each half of the transformer secondary is 100V. The load resistance is 975 Ω Find

(i) Average current

(ii) Average output voltage

(iii) PIV of diode.

GIVEN

Centre-tapped FWR, silicon diodes

V_{rms} (each half) = 100 V · R_L = 975 Ω · V_γ = 0.7 V (one diode conducts at a time)

↓

1 **Peak voltage of each half**

$V_m = \sqrt{2} \times 100 = \mathbf{141.42\ V}$

↓

2 **Peak load current**

One silicon diode is in the path:

$$I_m = \frac{V_m - V_\gamma}{R_L} = \frac{141.42 - 0.7}{975} = \frac{140.72}{975} = \mathbf{144.33\ mA}$$

3 **(i) Average (dc) current**

$$I_{dc} = \frac{2I_m}{\pi} = \frac{2 \times 144.33\ \text{mA}}{\pi} = \mathbf{91.88\ mA}$$

4 **(ii) Average output voltage**

$V_{dc} = I_{dc} \cdot R_L = 91.88\ \text{mA} \times 975\ \Omega = \mathbf{89.58\ V}$ (= 0.636 × 140.72)

5 **(iii) PIV of each diode**

The OFF diode sees the full secondary ($2V_m$) minus the drop of the ON diode:

$\text{PIV} = 2V_m - V_\gamma = 2(141.42) - 0.7 = \mathbf{282.1\ V}$ (ideal: $2V_m = 282.8$ V)

ANSWER

Results

(i) $I_{dc} \approx 91.9$ mA · (ii) $V_{dc} \approx 89.6$ V · (iii) PIV ≈ 282 V

If your class version also gives a diode forward resistance $r_f = 25\ \Omega$, use $I_m = 140.72/(975 + 25) = 140.72$ mA → $I_{dc} = 89.6$ mA, $V_{dc} = 87.3$ V. The method is identical.

Q9 Bridge rectifier, 220 V, 4 : 1, 200 Ω

QUESTION 9

A Bridge rectifier with ideal diodes has an ac source of RMS value 220 V, 50Hz connected to the primary of transformer. If the load resistance is 200Ω and turns ratio of transformer is 4:1, find the dc output voltage, dc output current.

GIVEN

Bridge rectifier, ideal diodes

Primary 220 V rms, 50 Hz · turns ratio $N_1 : N_2 = 4 : 1$ · $R_L = 200\ \Omega$

1 **Secondary rms voltage**

Step-down by the turns ratio:

$$V_{s(rms)} = \frac{220}{4} = \mathbf{55\ V}$$

2 **Secondary peak voltage**

$V_m = \sqrt{2} \times 55 = \mathbf{77.78\ V}$

3 **DC output voltage**

Ideal bridge:

$$V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 77.78}{\pi} = \mathbf{49.52\ V}$$

4 **DC output current**

$$I_{dc} = \frac{V_{dc}}{R_L} = \frac{49.52}{200} = \mathbf{247.6\ mA}$$

ANSWER

Results

$V_{dc} \approx 49.5\ V \cdot I_{dc} \approx 0.248\ A$

Q10 Bridge rectifier, 230 sin 314t, 10 : 1, 500 Ω

QUESTION 10

The input to a bridge rectifier is given through a 10 : 1 transformer from a supply of 230 Sin 314 t. If R_L = 500 Ω. Determine

DC load voltage

RMS load voltage

PIV across diodes

DC power delivered to the load

GIVEN

Bridge rectifier (diodes taken as ideal)

Primary $v = 230 \sin 314t \Rightarrow$ peak 230 V, $\omega = 314$ rad/s ($f = 314/2\pi = 50$ Hz) · 10 : 1 · $R_L = 500\ \Omega$

1 **Secondary peak voltage**

The given 230 V is already the **peak** (coefficient of sin), so just divide by the turns ratio:

$$V_m = \frac{230}{10} = \mathbf{23\ V}$$

2 **DC load voltage**

$$V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 23}{\pi} = \mathbf{14.64\ V}$$

3 **RMS load voltage**

$$V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{23}{1.414} = \mathbf{16.26\ V}$$

4 **PIV across each diode**

Bridge: PIV = V_m

PIV = **23 V**

5 **DC power delivered to load**

$$P_{dc} = \frac{V_{dc}^2}{R_L} = \frac{(14.64)^2}{500} = \mathbf{0.429\ W}\ (I_{dc} = 29.28\ mA)$$

ANSWER

Results

V_{dc} = 14.64 V · V_{rms} = 16.26 V · PIV = 23 V · P_{dc} ≈ 429 mW

If silicon diodes are assumed: V_{Lm} = 23 − 1.4 = 21.6 V → V_{dc} = 13.75 V, V_{rms} = 15.27 V, PIV = 23 − 0.7 = 22.3 V, P_{dc} = 0.378 W.